ALEXANDER FERRARA

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY **Atlanta, GA**

B.S. Computer Engineering | M.S. Computer Science

WORK EXPERIENCE

Booz | Allen | Hamilton **Remote**

Lead AI/ML Engineer | Generative AI April 2024 – Present

- **Architected and delivered multi-agent systems** used across internal and client environments, including a document co-authoring platform, an ITOps automation system for a major federal client, and reusable Agent-Oriented Architecture assets.
- **Designed, built, and deployed enterprise-scale, cloud-native AI solutions** on Azure and AWS, architecting full-stack applications with FastAPI/React and managing the end-to-end security posture of deployed services.
- **Self-managed risk and vulnerability mitigation for production environments**, resolving security findings flagged by Tenable scans and SVEs (System Vulnerability Evaluations) to maintain security compliance.
- **Engineered elastic architectures for high-volume workloads**, making strategic trade-offs between vertical/horizontal scaling and re-architecting monolithic containers into serverless fan-out (AWS Lambda/Azure Functions) to optimize throughput and cost.
- **Led teams of engineers** as Job Leader (5-8 engineers) driving architecture decisions, ensuring code quality, and providing technical mentorship as well as Career Manager (5 direct reports) supporting career growth, performance, compensation adjustments.
- **Modernized and scaled AI platforms**, leading redevelopment of a RAG-based Knowledge Assistant and Intelligent Data Processor (GraphRAG pipelines for entity extraction, disambiguation, linking, and intelligent retrieval).
- **Executed technical sales and strategy initiatives**, leading architectural workshops for executive stakeholders to translate complex AI capabilities into production roadmaps and successfully securing internal R&D project funding.
- **Anchored technical delivery** for distributed R&D teams, providing hands-on guidance to junior developers navigating complex GenAI codebases and resolving high-priority technical blockers to maintain team velocity.

Cruise **Remote**

Senior Software Engineer – Embedded Systems, Manufacturing June 2023 – February 2024

- **Led software enhancements** for manufacturing and health verification of autonomous systems across early development, end-of-line testing, and vehicle service.
- **Developed software in Python, C++, and ROS** (Robot Operating System) for data collection, analysis, and process optimization.

General Motors **Detroit, MI / Remote**

Senior Design Release Engineer – Autonomous Vehicle Computing December 2021 – June 2023

- **Provided technical leadership and decision-making** for functional hardware development teams resolving top pressing issues efficiently.
- **Built data processing tools** to analyze large datasets, identifying anomalies during manufacturing and forecasting device reliability, directly informing millions in spend decisions and enabling earlier logistics planning.
- **Applied knowledge gained from autonomous vehicle compute platform design and development** to optimize future compute systems development processes throughout full product development lifecycle.

Design Release Engineer – Autonomous Driving System Computer June 2019 – November 2021

- **Launched** a new autonomous vehicle computing hardware platform
- **Directed work of multiple international suppliers and cross functional team** efforts including senior engineers and SMEs.
- **Led project activities through design and development lifecycle** from early experimental validation through commercial launch.
- **Identified design, manufacturing, and assembly opportunities** to satisfy vehicle program needs while minimizing cost.
- **Bridged start-up agility with GM enterprise processes**, enabling rapid iteration while meeting strict automotive standards.
- **Negotiated and managed spend** on multimillion-dollar hardware/software development contracts.

Systems Engineer – Connectivity Technologies November 2018 – May 2019

- **Supported subject matter experts** for technologies including Apple CarPlay, Android Auto, Bluetooth, and Wi-Fi.
- **Led "Advanced Technology Work" project** centered on vehicle data offloading and secure Wi-Fi enablers for V2X applications.
- **Defined system specifications and requirements** for multiple functional modules.
- **Coordinated projection technology certifications** for LG Low Radio Program, collaborating with Apple and Google on infotainment program approvals.

Infotainment Execution Engineer July 2017 – October 2018

- **Delivered infotainment execution efforts** across ten vehicle programs throughout development lifecycle.
- **Developed software tools** to enhance workflow processes among infotainment and greater electrical community.
- **Conducted Level 1 infotainment support** at Proving Grounds facility providing issue root cause and resolution to ensure launch quality.

Booz | Allen | Hamilton — Atlanta, GA

Developer Intern — May 2014 – August 2016

Completed work full-time during summer semesters, as well as half-time throughout primary academic semesters

- **Contributed source code**, primarily focused on full stack web technologies, on eight client applications (federally contracted).
- **Developed for projects throughout each stage of the software performance lifecycle** (EPLC), all utilizing agile methodology.
- **Exemplified leadership and presentation skills** through management consulting-oriented intern case competition over the duration of each summer, winning first place while acting in the role of team leader.

Georgia Institute of Technology — Atlanta, GA

Undergraduate Teaching Assistant - Digital Design Laboratory — August 2015 – December 2015

Facilitated course focused on rapid prototyping of digital systems and FPGA design

Haplit – Georgia Tech Idea to Prototype Program, CreateX Startup Summer — August 2015 – May 2017

Multidisciplinary effort to develop a more affordable and durable learning aide for the visually impaired

Freelance Mobile Application Developer — November 2015 – August 2016

Independent contract development of a social networking mobile-fist application to connect athletes and organize events

SKILLS & INTERESTS

Core Expertise

- **Generative AI & Agentic Systems:** Multi-agent orchestration (LangGraph), Retrieval-Augmented Generation (RAG/GraphRAG), Document Co-authoring, and Intelligent Data Processing Systems.
- **Embedded & Automotive Systems:** Autonomous vehicle computing platforms, High Performance & Real-Time Architectures, kernel-level software optimization, C/C++, and robotics & sensor fusion.
- **Cloud & Infrastructure:** Azure (AI Search, OpenAI, Container Apps), AWS (ECS/Fargate, Bedrock, Neptune), Distributed Systems & Networking, Databricks, Docker, Kubernetes, and IaC (Terraform, Bicep).
- **Software Engineering & Leadership:** Python (primary), Data Structures & Advanced Algorithms, System & API Design, Technical Sales/Pre-sales Engineering, and Career & Project Leadership (8+ engineers).

Additional Experience

- **Development & Analytics:** Full-stack (FastAPI, React), Data Analysis & Modeling (SQL, MATLAB, simulation)
- **Technical Foundations:** Digital logic design (VHDL), FPGA prototyping, C#, Java, and .NET.

Achievements: GM Executive Reverse Mentorship Program, GM Vehicle Engineering TRACK Stars Fall 2018, DFSS Black Belt Certification

Hobbies: Skiing, travel, outdoor recreation, software development passion projects, 3D printing (esp when overlapping with SW element!)

RELEVANT COURSEWORK

Undergraduate Studies:

Embedded Computing Systems
Programming Hardware/Software Systems
Architecture, Concurrency, Energy in Computation
Integrated Circuit Fabrication
Capstone Design – Autonomous Boating

Graduate Studies:

High Performance Computer Architecture
Machine Learning for Trading
Artificial Intelligence for Robotics
Computer Networks
Algorithms